robot04 Mapping Experiments: 20 October 2026



BCIT Centre of Welding Technologies and Metallurgy Research

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Motion and Power Electronics Control (MPEC) Lab

18 autonomous exploration runs · Hector SLAM, Nav2 and frontier exploration on a Raspberry Pi 5 · robot04 (MentorPi T1)

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The Day at a Glance

20 October 2026, 09:32 – 12:57

18	153 m	32.5 min	8
autonomous runs	driven by the robot	explorer time	maps saved on the PC

1

Bounded area: works

7 of 9 enclosure runs finished and drove home; the last 6 runs had no failed goal

2

Large open area: not yet

82 % of goals aborted by Nav2 in the morning; no run got back to its start

3

Two side findings

battery check trips on voltage dips; new app timer matches the explorer within 12 s

No code changed during the day: the runs differ in where they ran and how they ended.

Setup and What Was Measured

Same software and parameters in every run

On the robot (Raspberry Pi 5)

Hector SLAM, 0.05 m cells, MS200 lidar 10 Hz

Nav2: NavFn + Regulated Pure Pursuit, 0.10 m/s

Frontier explorer: radius 0.15 m, min frontier 0.30 m, replan 3 s, no-progress 30 s, battery ≥ 7.0 V

On the PC (robot04 mapper app)

Watches map, path, Nav2 plan and frontiers

Starts the stack over ssh; Explore / Pause / Home

Saves pgm + yaml + json; new mapping timer saved with the map

Measure	Source	Definition
duration	explorer log	Explore pressed → done or operator stop
distance	explorer log	sum of SLAM pose steps
mapped area	map	known cells (free + occupied) × 0.05^2 m², also beyond walls
goals	explorer log	reached / frontier seen on the way / failed (Nav2 aborted, timeout)

Data: 18 explorer logs, 17 robot maps, 8 PC maps, all in experiments/2026-10-20/data.

Five Sessions Through the Day

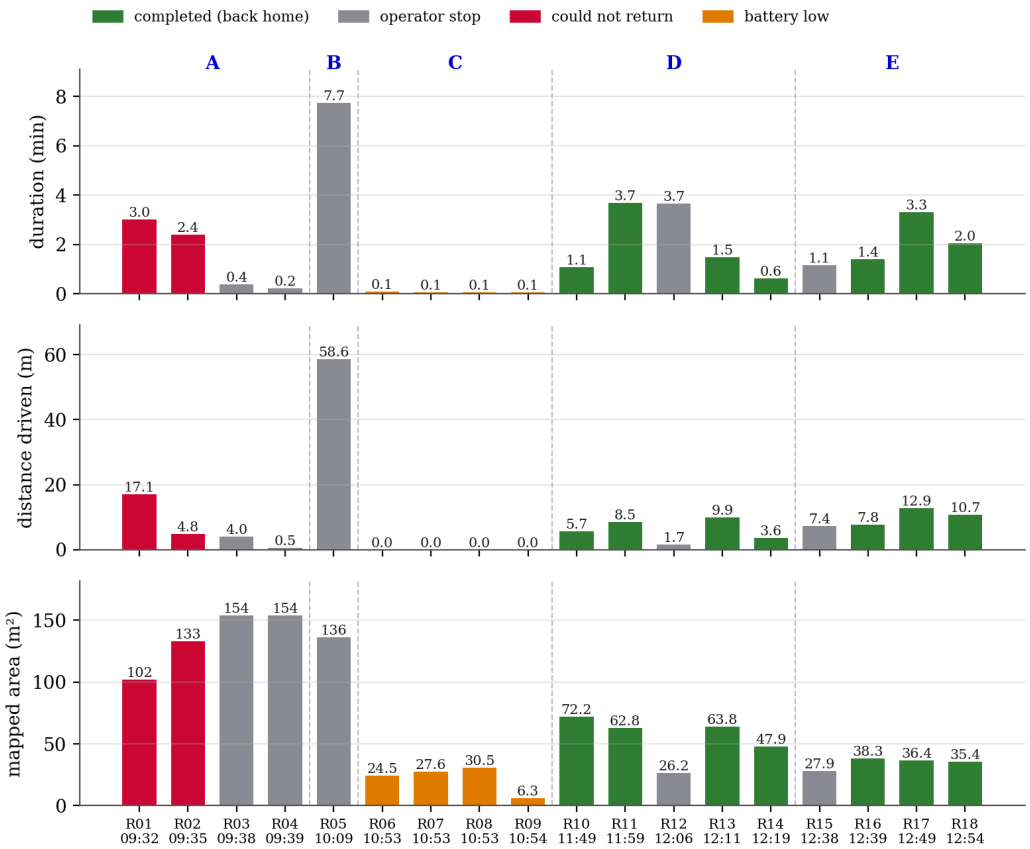
How each group of runs ended

A	B	C	D	E
09:32	**10:09**	**10:53**	**11:49**	**12:38**
Large open area	**One long run**	**Battery low**	**Small enclosure**	**Enclosure, repeat**
4 runs	**1** runs	**4** runs	**5** runs	**4** runs
0 back home	**0** back home	**0** back home	**4** back home	**3** back home
65 / 79 goals failed	**2 / 16** goals failed	**0 / 5** goals failed	**10 / 34** goals failed	**0 / 31** goals failed
26 m driven	**59 m** driven	**0 m** driven	**29 m** driven	**39 m** driven

Failures fell from 82 % (A) to 29 % (D) to 0 % (E) as the area became smaller and closed.

Every Run

Duration, distance and mapped area, coloured by outcome



Longest

R05: 7.7 min, 58.6 m, 136 m²

Completed

7 runs, 0.6–3.7 min

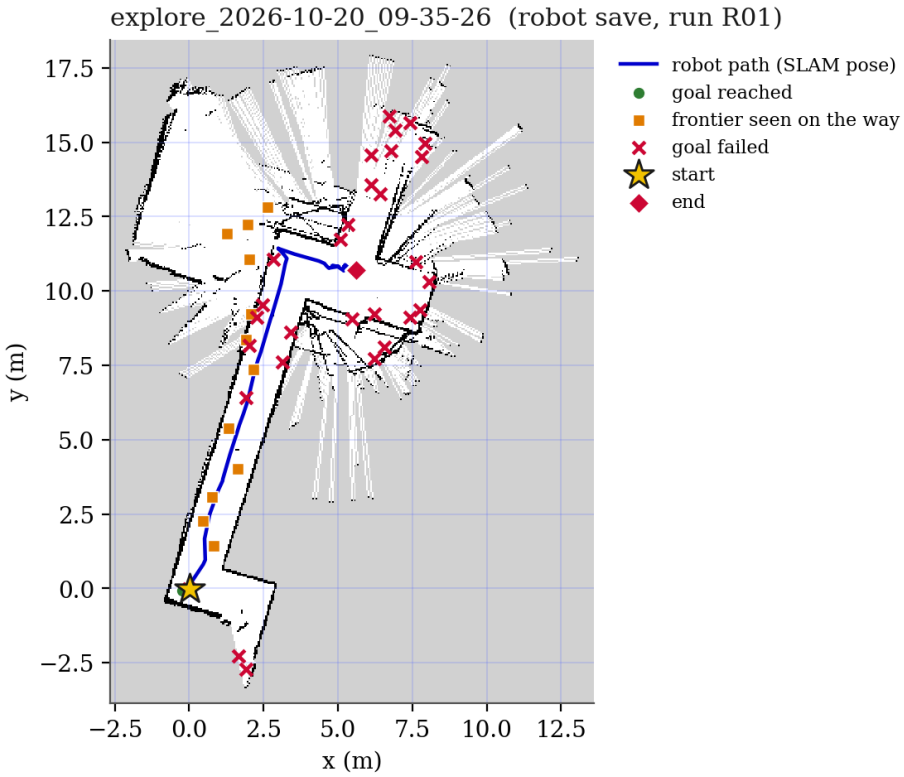
Cumulative

A and C kept one SLAM map across runs, so their area adds up

Green = explored to the last frontier and drove home without help.

Session A: Large Open Area

Nav2 aborted 62 of 74 goals in R01 and R02



29 / 41

R01 goals failed / sent

33 / 33

R02 goals failed / sent

Failed goals (×) sit at the far edge of the map and behind walls

Neither run could drive back to its start

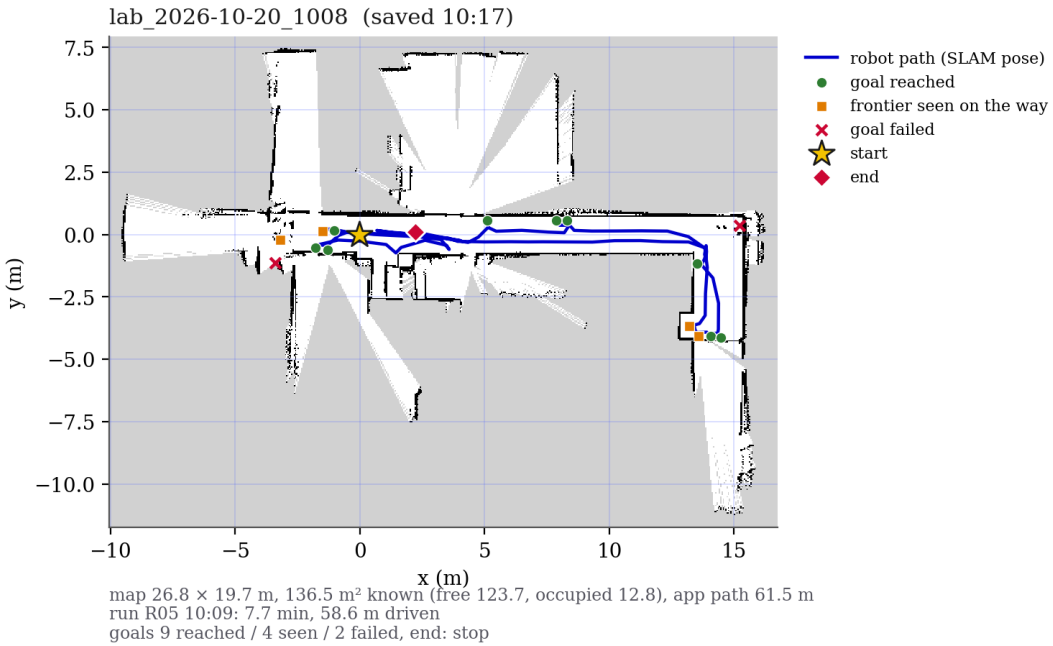
Frontier search 80–150 ms per goal here (20–70 ms in the enclosure)

Open question: why Nav2 aborts. Check the 0.45 m wall clearance in narrow passages and goals inside the inflated costmap.

The explorer kept choosing goals Nav2 could not plan to: 41 goals in 3 min in R01.

Session B: The Longest Run

R05, 10:09, saved as lab_2026-10-20_1008



map 26.8 × 19.7 m, 136.5 m² known (free 123.7, occupied 12.8), app path 61.5 m
run R05 10:09: 7.7 min, 58.6 m driven
goals 9 reached / 4 seen / 2 failed, end: stop

7.7 min
explored

58.6 m
driven

136 m²
mapped

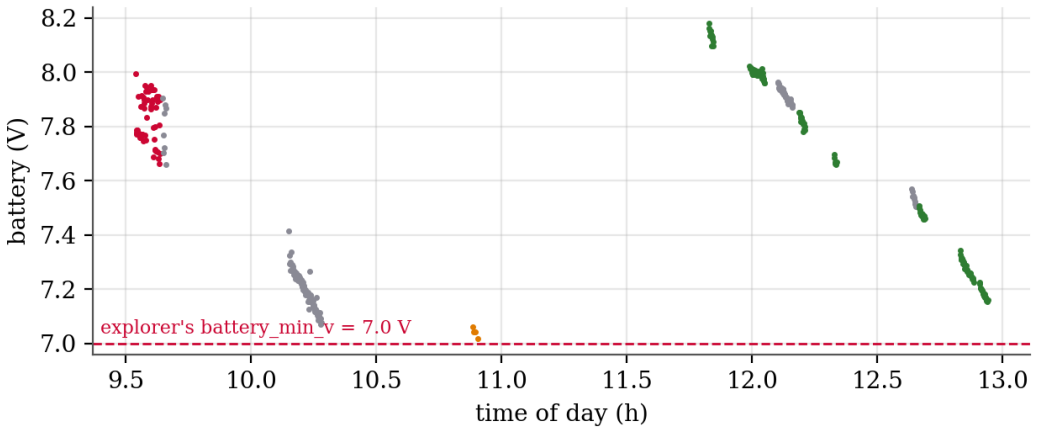
9 / 2
goals reached / failed

A corridor about 15 m long and the branch at its far end; stopped by the operator before it finished

In the same area as session A, one run reached 9 goals and drove 59 m.

Session C: The Battery Rule Fired at Once

Four runs at 10:53, each ended within 4 s



Run	Battery	Ran
R06	7.06 V	4 s
R07	7.04 V	4 s
R08	7.04 V	4 s
R09	7.02 V	4 s

Explorer limit: battery_min_v = 7.0 V

Logged samples 7.02–7.06 V, but the check uses the instantaneous voltage, which dips when the motors start

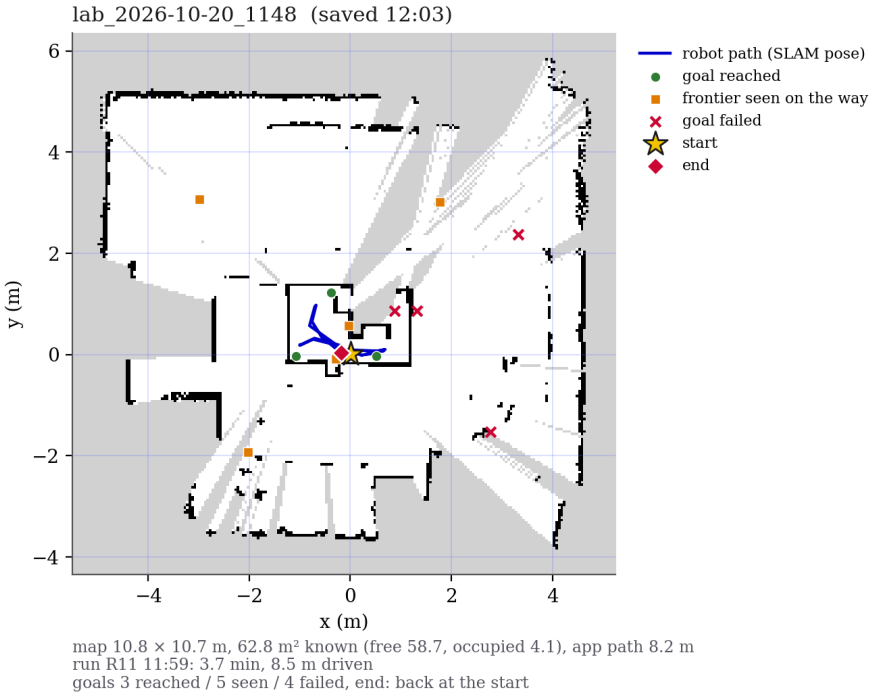
R05 had ended at 7.07 V; by 11:49 the battery read 8.18 V

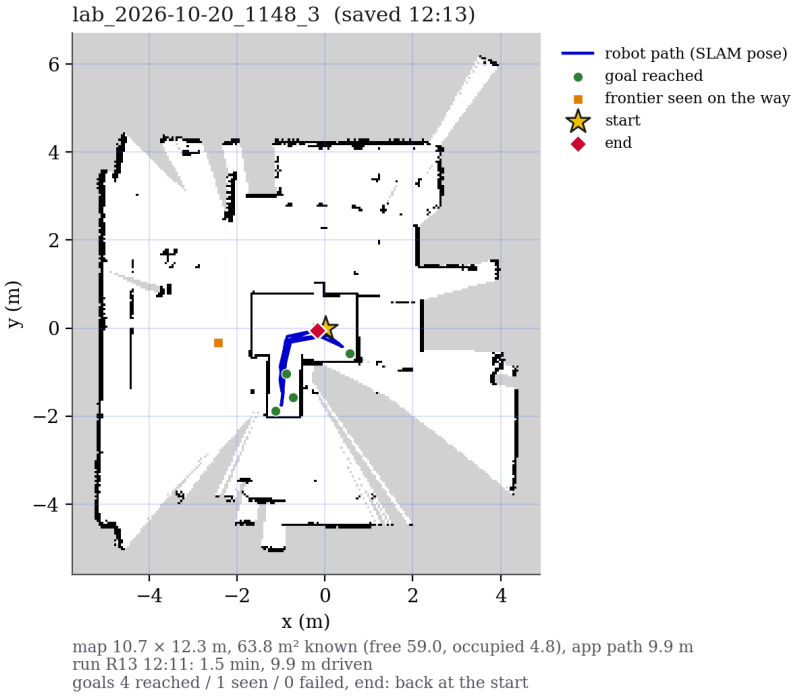
Fix: filter the voltage (e.g. a 5 s mean) before comparing it with 7.0 V, and log the minimum voltage.

A safety rule on a raw voltage stops the robot at the first load step.

Session D: A Small Enclosure

11:49 – 12:21, four of five runs finished on their own



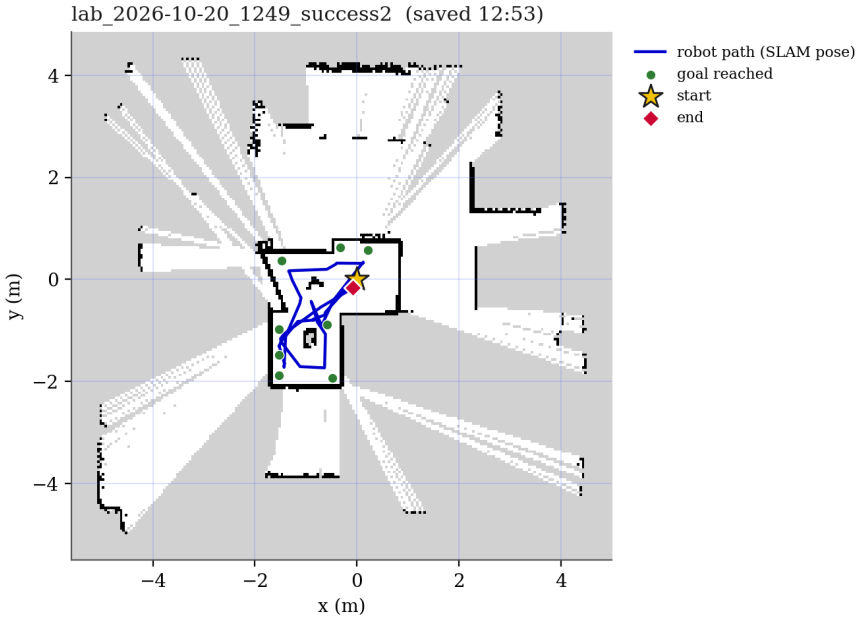


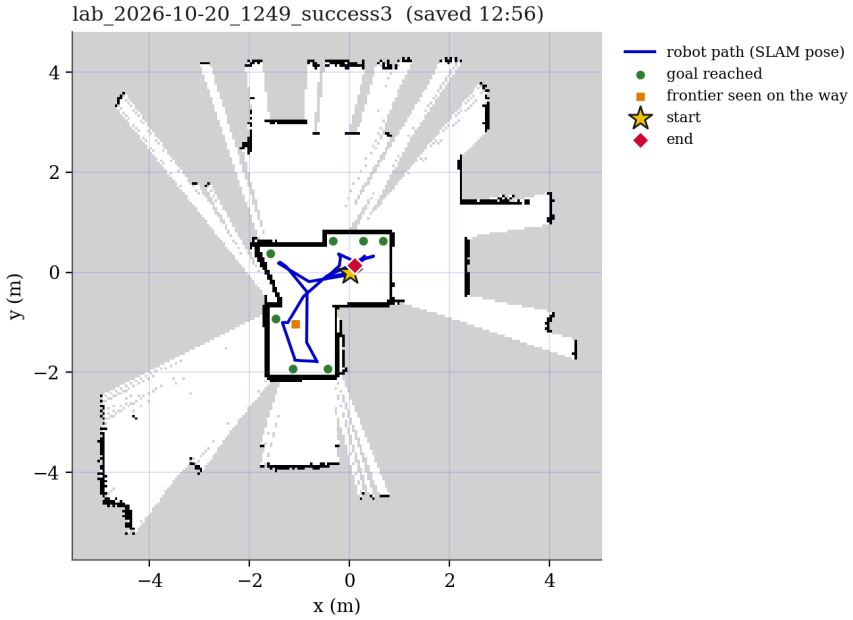
R11 (left): R11 11:59: 3.7 min, 8.5 m, 3 reached / 4 failed; failed goals outside the walls. **R13 (right):** R13 12:11: 1.5 min, 9.9 m, 4 reached / 0 failed.

The lidar sees out of the enclosure; goals placed out there cannot be reached and fail.

Session E: Repeat Runs, No Failed Goals

12:38 – 12:57, 31 goals sent, none failed



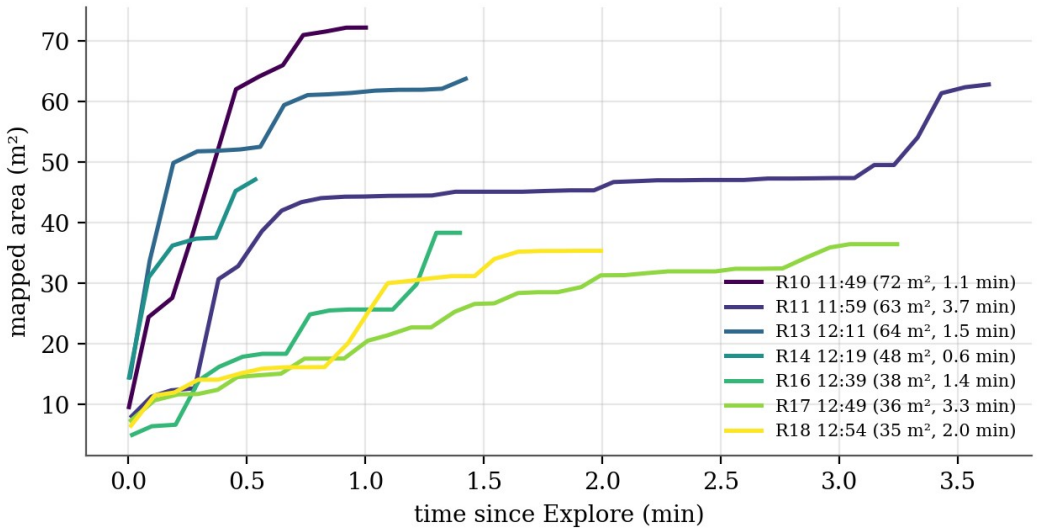


Completed runs R16–R18: 1.4–3.3 min, 7.8–12.9 m driven, mean 0.08 m/s (cruise 0.10), 5–14 s to drive home after the last frontier.

R17 (left) reached all 8 of its goals; R18 (right) 7, and saw 1 more on the way.

How Long Mapping Takes

Mapped area against time, and the new app timer



Steep start: each goal opens a new view for a 360° lidar

Then corners one at a time: R11 sat at 45–47 m² for more than 2 min

Completed runs: 0.6–3.7 min

Map	App timer	Explorer
R17	211 s	199 s
R18	130 s	123 s

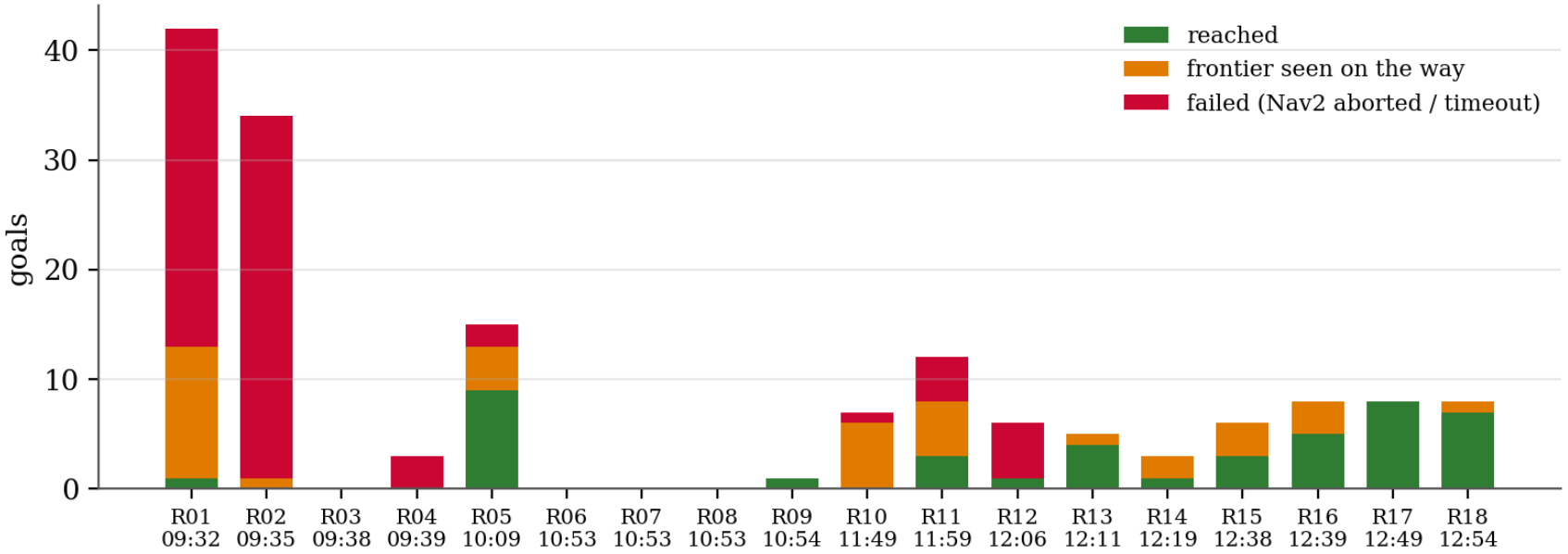
Timer runs while SLAM poses arrive and is saved in the map's json.

Reads 6–12 s longer: it also counts SLAM before Explore and the time until the map is saved.

Use the explorer time to compare runs; the app timer records the whole session.

Where the Goals Went

Goal outcomes per run



Session	A	B	C	D	E
failed / sent	65 / 79	2 / 16	0 / 5	10 / 34	0 / 31

Orange: frontiers the lidar cleared before the robot got there, so the explorer moved on.

The Saved Maps

~/Desktop/robot04_maps, lab_2026-10-20_*

Map	Run	Extent (m)	Known m²	Occupied m²	Path (m)	Timer
1008	R05	26.8 × 19.7	136.5	12.8	61.5	–
1148	R11	10.8 × 10.7	62.8	4.1	8.2	–
1148_2	R12	10.2 × 10.6	26.2	1.5	1.7	–
1148_3	R13	10.7 × 12.3	63.8	4.8	9.9	–
1148_4	R14	9.8 × 10.9	47.8	3.4	3.5	–
1237_success	R16	10.6 × 11.2	38.3	2.4	7.7	–
1249_success2	R17	10.6 × 10.3	36.4	2.6	12.8	211 s
1249_success3	R18	10.6 × 10.6	35.3	2.6	10.3	130 s

Every map is also plotted with metric axes, path and goals in figures/maps.

Findings and Next Steps

What to change before the next session

1. **Works in a bounded area:** 7 of 9 enclosure runs finished and drove home; last 6 runs: 0 failed goals
2. **Open areas: Nav2 aborts:** log the abort reason and costmap per failed goal; try a smaller inflation
3. **Unreachable frontiers:** drop frontiers with no path in the global costmap before sending them
4. **Battery rule:** filter the voltage before the 7.0 V check; log the minimum
5. **Timing:** explorer time per run; app timer saved with every map from now on

Next session: the same enclosure test, then the open area with Nav2 abort logging.